# Chapter 0, on its own

Every rule in one line, three proposed rules, and what does not fit. 13 November 2026.

Source: `rules.tex` on `main`, unchanged. Nothing here is adopted until E says so.

## Contents


**A. The rules, one line each** 1

**B. Proposed, awaiting E** 4

**C. Duplicates, gaps, and the buildings** 5

**D. Chapter 0 as it stands** 6

**Chapter 0. The rules** 6


## A. The rules, one line each

In the order of `rules.tex`. The full wording is Part D.

### Background

**S1** A bag is a set; sets, maps and logic as usual.

**S2** A floor: a bag $X$, operations $O$, questions $Q$.

**S3** A law is a statement about a floor; an equation is "$s = t$ for all letters", made from letters and operations.

### Build: six ways, nothing else

**M1** Pairs $X \times Y$; lists $X^n$.

**M2** Sequences $\mathbb{N} \to X$.

**M3** Words from names with numbers of inputs; equal only if written the same; maps out by recursion.

**M4** Select: the $x$ in $X$ meeting a condition.

**M5** Bags of a blur: $X/\sim$.

**M7** Sub-bags: all sub-bags of $X$. Far larger; say so. Maps and blurs are bags by M1, M7, M4.

**M6** Lifting: an operation goes up place by place, or by a formula in a Definition; a selected bag keeps it only if closed.

## Blur

**L1** Blur: $x \sim x$, runs back, chains. Bag $[x]$.

**L2** Respects $f$: glued in, glued out.

**L3** Keeps $q$: glued $\Rightarrow$ same answer (one thing or two).

**L4** Blurrier: glues at least what the other glues.

**L5** From below: least blur gluing $G$, respecting $O$.

**L6** From above: blurriest blur keeping $Q$, respecting $O$.

**L7** Every Blur line names its $O$; other $O$, other blur.

**L8** One-way blur: $x \preceq x$, chains, respects as L2; its two-way part is a blur, ordered by $\preceq$.

**L9** Likeness: $x\ N\ x$, runs back, need not chain.

**L10** Limits: $O$ holds finite-input operations only; a blur may respect limits for a stated nearness, in a slot *Limits:*; no limit and no question in $O$.

## Theorems about blurs

**Z1** Equality is least, one bag is blurriest; both respect everything.

**Z2** From below exists, by a process: close $G$ up under back, $O$, chain; repeat.

**Z3** From above exists: glue $x, y$ iff $q(c(x)) = q(c(y))$ for all $q$ in $Q$ and all contexts $c$.

**Z4** Sandwich: gluings kept in every context $\Rightarrow$ below no blurrier than above.

**Z5** Appears: below = above once every unglued pair is split by one context and one question.

**Z6** Descent: respected operations and kept questions go down to $X/\!\sim$.

**Z7** Equations are free: on $X^n$, sequences, closed selections, $X/\!\sim$.

**Z8** Every other law is checked again, or its loss recorded.

**Z9** Normal form $\nu$: $\nu\nu = \nu$; one fixed member per bag.

**Z10** A likeness is a blur seen through a picture; every likeness arises so.

## Arrows

**W1** Arrow: a map between floors, with the list of what it respects.

**W2** Inside: one-to-one. Picture: reaches all.

**W3** $x \mapsto [x]$ is a picture.

**W4** Every arrow is its own blur, then an inside.

### Kinds of chapter

**K1** *blur*: one new floor $X/\sim$; slots of the table below.

**K2** *build*: new bag by M1–M7, Blur "none"; carried / not carried.

**K3** *select*: Tests; At most one; At least one (Build, or "closed"); Read off.

**K4** *arrow*: the arrow; respects and dies; its blur; inside or picture.

**K5** *count*: what is counted, the count, why it forces; Theorems only.

**K6** *table*: rows, tests, entries; read-offs are conjectures.

**K7** *rerun*: earlier chapter, new Given; free (Z7) and checked again (Z8).

### Slots of a blur chapter

A or G (Given) · D (Definitions: Build, lift, questions) · B1 (from below: $G$, $O$) · B2 (from above: $Q$, same $O$) · P (Bags, Normal forms, Survives, Dies, Appears, Arrow) · N (Notation, only after the blur) · T (Theorems) · R (Remarks) · E (Exercises).

### Conventions

**C1** Every Given cites the earlier line that proves it (Chapter 1 axioms apart).

**C2** Survives and Dies name only what is defined on the floor blurred.

**C3** A reused sign: lift, survive, descend, agree.

**C4** Dies follows from the Blur line; write only the traps.

**C5** Blur "none": Survives/Dies become carried/not carried.

**C6** The first choice of gluings or questions is open; the rest is forced.

**C7** Reserved words: Survives (L2, L3 only); "kept on those bags" is a Theorem; glued never means near.

**C8** Imported: a Given "(Imported: source.) statement. Used by: lines.", listed in the building's ledger; unused classics are "reported".

**C9** Made from above: if the only gluings are the question's own, write "B1: none (made from above)" and claim no Appears.

### The second floor

**F1** Notation gives a system, Arrow an arrow, Theorems yes/no facts.

**F2** Test: a law askable of every system. Table: systems × tests.

**F3** Blur on a table: same answers on the tests $U$.

**F4** Class: the rows passing the tests $T$.

**F5** Closed: running the chapter on the system gives an inside and a picture.

**F6** The table suggests; proof or a new row decides.

**F7** Floor two reads floor one, never back.

**F8** Loop: build $X$, blur by an earlier $S$; $S$ comes back closed, changed or wild; a theorem of the later chapter.

### Remarks (not rules)

**R1** S1, M1–M5, M7 are a small part of set theory.

**R2** Below makes, above justifies; Appears is their agreement.

**R3** On $\mathbb{R}^n$ the choice of $O$ separates four roads.

**R4** "No zero divisors" is lost by Blur ($\mathbb{Z}/6$) and by Build ($\mathbb{R}^2$).

**R5** Old names.

**R6** A rule relies on nothing after Chapter 0; so Amplify is not a rule (Field notes, Note 2, line A). F8 stays.

**Count.** Background 3, Build 7, Blur 10, Theorems 10, Arrows 4, Kinds 7, Conventions 9, Second floor 8: **58 rules**, plus 6 remarks and 4 exercises. Retired: Z11 (Amplify, now Field notes line A). There is no Z12 in Chapter 0 or in any building.

## B. Proposed, awaiting E

Proposed today for stating most of mathematics (`plans/BUILD-ORDER-MOST.md`, Stage 0). **Not adopted.** Labels X1–X3 are temporary. For each: a draft line in the style of Part A, what it unlocks, and whether the existing rules already make it.

**X1** (Chosen.) From a bag of non-empty bags: one member of each. Written as a Given: *(Chosen: one at a time / a sequence / any bag of bags.) statement. Used by: lines.* Listed in the ledger beside Imported lines (C8).

*Unlocks:* Zorn, Hahn–Banach, a basis of every vector space, Tychonoff.
*Made from existing rules?* No: no line of M1–M7 picks.
*Already needed now:* Chapter 26 picks one cover for each $N_i$ (its "honest note"), and Chapter 27, T1 picks "one continuous member, rung $n$ for the $n$-th". Both need a sequence of choices only. Chapter 27, R3 and Chapter II.3, R4 say outright that a rule of choice is missing (C, items 13–14).
*Fit with the method:* a chosen member is never a normal form (Z9) and never gives Appears; blur proves "at most one", choice gives "at least one" with no recipe. Natural home: a third way to fill K3's *At least one* (Build / closed / Chosen).
*For E:* one strength or three? (a sequence of choices covers Chapters 26–27; full choice is needed for Zorn and Hahn–Banach).

**X2** (Charts.) Given open pieces $U_i$ of $\mathbb{R}^n$ ($i$ in a bag $I$) and change maps $\varphi_{ji}$ of a stated kind (continuous, smooth, $\mathbb{C}$-straight) with $\varphi_{ii} = \mathrm{id}$ and $\varphi_{kj}\varphi_{ji} = \varphi_{ki}$: the bag glued from the pieces along the changes.

*Unlocks:* manifolds, Riemann surfaces, varieties, bundles.
*Made from existing rules?* Very nearly, and in the blur's own way: build the pairs $(i, x)$ with $x$ in $U_i$ (M1, M4); blur from below by $(i, x) \sim (j, \varphi_{ji}(x))$ (L5), respecting nothing, *Limits:* the nearness of $\mathbb{R}^n$ (L10). What is new is only (a) the word "kind" for the change maps

and (b) one theorem: the two conditions on $\varphi$ make each bag meet each piece at most once. "Two points never pulled apart" (the line with two origins) is a Dies line, not automatic. *For E:* a new Build line, or a convention plus one theorem about L5?

**X3** (Countable closure.) From $X$ and a list of sub-bags: the smallest bag of sub-bags of $X$ holding them and closed under complement and under the union of a sequence.

*Unlocks:* measure, probability, Borel sets.
*Made from existing rules?* Yes, as a select: among the sub-bags of the sub-bags of $X$ (M7 twice), take the meet of all families that hold the list and are closed (M4). It is a from-above object: there is no process (a process needs stages past every sequence, and ordinals are not built). Chapter 26 already uses unions of sequences of covers.
*For E:* a new line, or a named select (K3) with its cost (M7 twice) said openly?

## C. Duplicates, gaps, and the buildings

Each item: what, where, and the choices. Merge decisions are E's. Sources: `rules.tex`, all part files of Buildings 0–2 on `main` (13 Nov).

### Inside Chapter 0

**1** (Order of Build.) "Six ways" are M1–M5, M7; M6 (lifting) sits after M7 in its own list, and is not a way to build. M2 is a case of M7, M1, M4, as M7 itself says. *Choices:* renumber (M6 $\leftrightarrow$ M7); M2 to a remark; leave.

**2** (Likeness twice.) L9 defines likeness as a rule; Z10 says it "is not a new primitive". *Choices:* keep L9 as a definition only; fold it into Z10.

**3** (Duplicates.) C5 = K2 (carried / not carried). C1 and C8 are both about Given lines. The table row *Survives* restates C2 and C7. C7's last sentence (a theorem with a hypothesis is a Theorem) restates its second. *Choices:* merge C5 into K2; one "Given" rule from C1 + C8.

**4** ("Closed" has four meanings.) M6: closed under an operation. K3: "closed" as the reason for *At least one* (complete). F5: a system on which the chapter returns an inside and a picture. F8: the verdict "closed". *Choices:* keep one, rename three.

**5** ("Carried" has two.) C5/K2 reserve it for build chapters, but about 25 Notation lines in all three buildings say "carried over (Z6)" for C3's *descend*. *Choices:* write "descended" in Notation lines; or widen C5.

**6** (Rules that point ahead.) L10 needs the nearness of Chapter 5, and C7 cites Chapter 5, P5, T1; R6 says a rule relies on nothing after Chapter 0. *Choices:* state nearness in Background (a ladder of likenesses, as Chapter 5, D4); or allow the forward reference as a reading convention, like F8.

**7** (Missing existence lines.) Z3 handles questions of one thing; L3 and L6 also allow questions of two things. Z2, Z3 give existence for L5, L6 only: nothing for a one-way blur from below or above (L8), nor for a blur that respects limits (L10), which Chapter 26, B1 uses. Z2's "repeat until nothing new" needs "take the union of all stages" when $X$ is infinite. Each is one line (the meet of all such relations exists). *Choice:* add them to Z2–Z3, or leave.

**8** (Table vs C9.) The slot table says Appears is "B1 = B2, by Z4, Z5"; C9 removes Appears when B1 is none. Only C9 says so. *Choice:* one clause in the table.

**9** (Opening line.) "Every later chapter uses these rules and adds none." True today; X1–X3 would add the first new rules since F8.

### How the buildings use the rules

**10** (L7: Blur lines without $O$.) Building 1: the B1 lines of Ch1, Ch2, II.2, II.3, II.4, II.5, II.7, II.8, II.9, II.10, II.13, III.1 give gluings but no *Respects:*. Building 0, Chapter 6, B1 ($x \sim_W y \iff x - y \in W$) has neither *From below* nor $O$. Building 0 elsewhere and Building 2 comply.

**11** (C9: every glued pair listed as "from below".) Chapter 17, B1 glues $a \sim b$ whenever $c\,a_n \le b_n \le K a_n$: that is the whole blur. Building 2, Ch1, B1 ($A \sim B$ when $\vdash A \leftrightarrow B$) and Ch2, B1 ($\vdash \Box A \leftrightarrow \Box B$): each list is already the blur. Building 1, II.1 calls the table blur "B1", while Building 0, II.1 writes "B1: None (made from above, C9)".

**12** (Kinds line: "these words and no others".) Chapters 7 and 14 write "build (words)", Chapter 10 "build (sub-bags)". Chapter II.2 is marked *blur* though it reruns I.3, and II.5 calls itself "a rerun chapter, in the style of II.2".

**13** (Choice used with no rule.) Chapter 26 (one cover per $N_i$) and Chapter 27, T1 (one member from each bag, the $n$-th at rung $n$). Not in either ledger. This is a gap today, whatever is decided on X1.

**14** (Choice called for.) Chapter 27, R3 (sets with no length) and II.3, R4 (a basis of $C$) say "needs a rule of choice (not in Chapter 0)".

**15** (Stale citation.) Building 1, Ch1, G1 cites "blur L1–L9"; L10 came later.

**16** (Shadowed labels.) Chapter 32 labels its earned Stokes lines S1–S4, the same labels as Background S1–S3. Cosmetic; `build.py` does not catch it.

## D. Chapter 0 as it stands

`rules.tex`, unchanged, as it appears in every building.

### Chapter 0. The rules

What may be built, what a blur is, and what each line of a chapter claims.

This chapter is the only place where the method itself is stated. Every later chapter uses these rules and adds none.

### Background

**S1** A *bag* is a set. We use sets, membership, subsets and maps in the ordinary way, and ordinary logic.

**S2** A *floor* is a bag $X$ together with a list $O$ of *operations* (maps $X \times \cdots \times X \to X$) and a list $Q$ of *questions* (maps from $X$, or from $X \times X$, to some bag of answers; a yes/no question is a question).

**S3** A *law* is a statement about a floor. A law is an *equation* if it has the form "$s = t$ for all values of the letters", where $s$ and $t$ are built from the letters and the operations only.

## Build

Six ways to make a new bag from bags already made. Nothing else may be built.

**M1** (Pairs and lists.) From $X$ and $Y$: the bag $X \times Y$ of pairs. From $X$ and $n$ in $\mathbb{N}$: the bag $X^n$ of lists of length $n$.

**M2** (Sequences.) From $X$: the bag of all maps $\mathbb{N} \to X$.

**M3** (Words.) From a list of names, each with a number of inputs: the smallest bag that holds $f(w_1, \ldots, w_k)$ whenever $f$ takes $k$ inputs and it holds $w_1, \ldots, w_k$. Two words are equal only if written the same. A map from the words to a floor is fixed by saying what it does to each name (definition by recursion).

**M4** (Select.) From $X$ and a condition: the bag of those $x$ in $X$ that meet the condition.

**M5** (Bags of bags.) From $X$ and a blur $\sim$ of $X$ (L1): the bag $X/\!\sim$ whose members are the bags of $\sim$.

**M7** (Sub-bags.) From $X$: the bag of all sub-bags of $X$.
A *map* $A \to X$ is a sub-bag of $A \times X$ in which each $a$ in $A$ stands on the left of exactly one pair. So the bag of all maps $A \to X$ is built by M1, M7, M4; M2 is the case $A = \mathbb{N}$. A blur of $X$ is a sub-bag of $X \times X$, so the bag of all blurs of $X$ is built the same way. M7 is the one line whose result is far larger than what went in; a chapter that uses it must say so.

**M6** (Lifting.) An operation of $X$ may be lifted to a built bag *place by place*; any other operation on a built bag must be given by a formula in a Definition line. A selected bag keeps an operation only if it is closed under it, and the chapter must say so.

## Blur

**L1** (Blur.) A *blur* of $X$ is a relation $\sim$ on $X$ with $x \sim x$; $x \sim y \Rightarrow y \sim x$; $x \sim y$, $y \sim z \Rightarrow x \sim z$. The *bag* of $x$ is $[x] := \{y : y \sim x\}$.

**L2** (Respects.) $\sim$ *respects* an operation $f$ if $x_1 \sim y_1, \ldots, x_k \sim y_k \Rightarrow f(x_1, \ldots, x_k) \sim f(y_1, \ldots, y_k)$.

**L3** (Keeps.) $\sim$ *keeps* a question $q$ if $x \sim y \Rightarrow q(x) = q(y)$. For a question of two things: $x \sim x'$, $y \sim y' \Rightarrow q(x, y) = q(x', y')$.

**L4** (Blurrier.) $\sim$ is *blurrier* than $\sim'$ if $x \sim' y \Rightarrow x \sim y$.

**L5** (From below.) Given a list $G$ of *gluings* (pairs to be glued) and a list $O$ of operations: the blur *forced by* $G$ *respecting* $O$ is the least blurry blur that glues every pair in $G$ and respects every operation in $O$.

**L6** (From above.) Given a list $Q$ of questions and a list $O$ of operations: the blur *kept by $Q$ respecting $O$* is the blurriest blur that keeps every question in $Q$ and respects every operation in $O$.

**L7** (A blur must say what it respects.) Every Blur line names its $O$. The same gluing with a different $O$ is a different blur.

**L8** (One-way blur.) A *one-way blur* of $X$ is a relation $\preceq$ with $x \preceq x$ and $x \preceq y,\ y \preceq z \Rightarrow x \preceq z$. It respects $f$ as in L2. Its *two-way part*, "$x \preceq y$ and $y \preceq x$", is a blur, and $\preceq$ is an order on the bags of that blur. A blur is a one-way blur that also runs backwards.

**L9** (Likeness.) A *likeness* on $X$ is a relation $N$ with $x\ N\ x$ and $x\ N\ y \Rightarrow y\ N\ x$. It need not chain. A blur is a likeness that chains.

**L10** (Limits.) $O$ holds only operations with finitely many inputs (S2). A blur may also be asked to *respect limits* for a stated nearness (Chapter 5): if $x_n \sim y_n$ for every $n$, $x_n$ comes near $x$ and $y_n$ comes near $y$, then $x \sim y$. The Blur line then has a slot *Limits:* naming the nearness. A limit is never put in $O$, and a question is never put in $O$.

### Theorems about blurs

Stated once here, used in every chapter.

**Z1** (Two ends.) Equality is the least blurry blur of $X$. "Everything in one bag" is the blurriest. Both respect every operation.

**Z2** (From below exists.) The blur of L5 exists. It is reached by a process: start from $G$; add $x \sim x$; add $y \sim x$ for each $x \sim y$; add $f(\dots x \dots) \sim f(\dots y \dots)$ for each $x \sim y$ and each $f$ in $O$; add $x \sim z$ for each chain $x \sim y \sim z$; repeat until nothing new appears.

**Z3** (From above exists.) The blur of L6 exists. For questions of one thing, it glues $x$ and $y$ exactly when $q(c(x)) = q(c(y))$ for every $q$ in $Q$ and every *context* $c$: a map $X \to X$ made from the operations in $O$ and fixed members of $X$ (for example $c(x) = x + z$).

**Z4** (Sandwich.) If every gluing in $G$ is kept by every question in $Q$ in every context, then the blur from below is no blurrier than the blur from above.

**Z5** (Recipe for Appears.) In Z4 the two blurs are *equal* as soon as: for every $x \not\sim y$ (from below) there is a context $c$ and a $q$ in $Q$ with $q(c(x)) \neq q(c(y))$.

**Z6** (Descent.) If $\sim$ respects $f$, then $f([x_1], \dots, [x_k]) := [f(x_1, \dots, x_k)]$ is an operation on $X/\!\sim$. If $\sim$ keeps $q$, then $q([x]) := q(x)$ is a question on $X/\!\sim$.

**Z7** (Equations are free.) An equation that holds on $X$ also holds: on $X^n$ and on sequences with operations lifted place by place; on any selected bag closed under the operations; and on $X/\!\sim$ for any blur respecting the operations.

**Z8** (Everything else must be checked.) A law that is not an equation (one that says "or", "$\neq$", "there is", or uses a question) may be lost by Build, Select or Blur. Each chapter must prove it again or record its loss.

**Z9** (Normal forms.) A *normal form* for $\sim$ is a map $\nu : X \to X$ with $\nu(x) \sim x$, and $x \sim y \Rightarrow \nu(x) = \nu(y)$. Then $\nu(\nu(x)) = \nu(x)$, and each bag holds exactly one $x$ with $\nu(x) = x$.

**Z10** (A likeness is a blur seen through a picture.) If $\sim$ is a blur of $X$ and $p : X \to Y$ is a picture, then "$y\ N\ y'$ iff $y = p(x)$, $y' = p(x')$ for some $x \sim x'$" is a likeness on $Y$, and it may fail to chain. Every likeness arises this way. So likeness is not a new primitive: it is made from blur and picture.

## Arrows

**W1** An *arrow* is a map $f : X \to Y$ between floors, with a list of the operations and questions it respects: $f(x + y) = f(x) + f(y)$, and so on.

**W2** An arrow is an *inside* if it is one-to-one, and a *picture* if it reaches every member of $Y$.

**W3** (Every blur is a picture.) $x \mapsto [x]$ is a picture $X \to X/\sim$ respecting whatever $\sim$ respects.

**W4** (Every arrow is a blur, then an inside.) For an arrow $f$, "$x \sim y \iff f(x) = f(y)$" is a blur of $X$ respecting what $f$ respects, and $f$ is the picture $X \to X/\sim$ followed by an inside $X/\sim \to Y$.

## Kinds of chapter

Every chapter is one of seven kinds, or a few of them in a stated order. The second line under a chapter title names the kinds, in order, with these words and no others. A chapter claims a blur only if it has a Blur line. In every kind, a name comes only after the line that makes or pins its object (a Notation line), old names go to the Remarks, and every sign reused from an earlier floor passes C3 in an Arrow line.

**K1** (*blur.*) Makes one new floor $X/\sim$. Slots: the table below.

**K2** (*build.*) Makes a new bag by M1–M7 and blurs nothing. The Blur line reads "none"; Survives and Dies become *carried* and *not carried* (C5).

**K3** (*select.*) Pins one object by what it must do. Slots: *Tests* (the conditions, a Select M4); *At most one*; *At least one* (a Build, or "closed", said once); *Read off* (laws proved from "only one"). A select may ask a question of a normal form of an earlier blur and find it kept on some bags (C7).

**K4** (*arrow.*) Reads one arrow between floors already made. Slots: the arrow; what it respects and what dies through it (W1); its own blur (W4); inside or picture (W2).

**K5** (*count.*) Existence from a count, said openly: what is counted, the count, and why the count forces the object. Its lines are Theorems, not Survives lines.

**K6** (*table.*) Rows, tests, entries (F2–F4); a survey of all arrows or all blurs of one sort is a table. What is read off is a conjecture until proved (F6).

**K7** (*rerun.*) Runs an earlier chapter with a new Given. Slots: *free* (equations, carried by Z7) and *checked again* (every other law, Z8), each line citing the line it reruns.

### The lines of a blur chapter

A blur chapter (K1) makes one new floor. Its lines, in order, and what each one claims:

| Label | Line | Claim |
|---|---|---|
| A or G | Axioms / Given | laws assumed (Chapter 1 only), or earned earlier and cited |
| D | Definitions | Build (M1–M5, M7), lifted operations (M6), questions. Adds no truth |
| B1 | Blur, from below | the gluings $G$ and the respected $O$ (L5) |
| B2 | Blur, from above | the questions $Q$ and the same $O$ (L6) |
| P | Bags | a test for $x \sim y$ that can be computed |
| P | Normal forms | a $\nu$ as in Z9 |
| P | Survives | $\sim$ respects an operation (L2) or keeps a question (L3) |
| P | Dies | $\sim$ does not keep a question: one witness $x \sim y$, $q(x) \neq q(y)$ |
| P | Appears | B1 = B2, proved by Z4 and Z5, naming the separating context |
| N | Notation | names for $X/\sim$ and its members. Only after the blur |
| P | Arrow | an arrow old $\to$ new, inside or picture, and what it respects |
| T | Theorems | statements inside the new floor, in the ordinary form |
| R | Remarks | other blurs, other roads, links to other chapters |
| E | Exercises | |

**C1** (Earned Given.) Apart from the axioms of Chapter 1, every Given line cites the line of an earlier chapter that proves it.

**C2** (Survives names only what is defined.) A Survives or Dies line may name only operations and questions defined on the floor being blurred.

**C3** (One sign, four steps.) A sign such as + may be reused on the new floor only after: *lift* (D), *survive* (P), *descend* (Z6), and *agree* (the Arrow respects it).

**C4** (Dies is for traps.) Every Dies line follows from the Blur line. Write only those a reader might wrongly use.

**C5** (Build-only chapters.) If the Blur line is "none", Survives and Dies are replaced by *carried* and *not carried*: which laws of the Given still hold on the built floor (Z7, Z8).

**C6** (The first choice is open.) Which gluings or questions to start from is the author's choice. Everything after that choice is forced, and is proved.

**C7** (Reserved words.) *Survives* names a blur and an operation it respects (L2) or a question it keeps (L3), and nothing else. A question that takes one answer on

every member of some bags, but not of all, is *kept on those bags*; such a line is a Theorem. *Glued* means "in one bag of a blur" and never "near" (Chapter 5, P5, T1: a blur that holds a rung of nearness has one bag). A theorem with a hypothesis is a Theorem, not a Survives line.

**C8** (Imported.) A step that a later line uses and that the books do not prove is written as a Given line of the form *(Imported: source.) statement. Used by: lines.* Each building lists its Imported lines in a ledger on its first page. A classical fact that no line uses stays in the Remarks, marked "reported".

**C9** (Made from above.) When the only gluings one could list are the pairs that the question already glues, the Blur line says "B1: none (made from above)", and the chapter makes no Appears claim. Listing $e \sim 0$ for each $e$ in the bag of 0, and letting the respected operations spread it (Z2), is a from-below line; listing every glued pair is not.

## The second floor

**F1** A chapter's Notation gives a *system*; its Arrow gives an arrow between systems; its Theorems give yes/no facts about the system.

**F2** A *test* is a law that can be asked of every system. A *table* has systems as rows, tests as columns, and the facts as entries.

**F3** (Blur on a table.) For a set $U$ of tests: $K \sim_U L$ iff $K$ and $L$ answer every test in $U$ alike.

**F4** (Class.) For a set $T$ of tests: class($T$) is the set of rows passing every test in $T$. A blur asks what tests cannot tell apart; a class asks what tests pick out.

**F5** (Closed.) A system is *closed* for a chapter if running the chapter on it gives an Arrow that is both an inside and a picture.

**F6** (The table suggests; rows refute.) A statement read off a table is a conjecture. It becomes a theorem by proof from the tests, or dies when a row is built that breaks it.

**F7** (One way only.) Floor two reads floor one, never the reverse. A first-floor chapter cites only first-floor chapters; a class of the second floor only names what its rows already have. A Remark may point ahead, but no line may rely on what it points to.

**F8** (Loop.) A chapter may build a system $X$ and blur it by the action of an earlier system $S$. A *loop* is a theorem of that chapter saying that $S$ comes back: a question on the bags whose answers match $S$, respecting the operations named. Verdicts: *closed* ($S$ comes back exactly), *changed* (larger or smaller), *wild* (no rung of nearness keeps the bags apart; then keep the moves, do not forget them). The return is a theorem of the later chapter: by F7 no earlier line may rely on it.

## Remarks

**R1** (What is assumed.) S1, M1–M5 and M7 are a small part of ordinary set theory. The book does not replace that foundation; it organises what is built on it.

**R2** (Why two blurs.) From below is how a blur is *made*: it tells you what is glued. From above is why it is *right*: it tells you what the gluing is for. Appears is the claim that the two agree.

**R3** (Why L7.) On $\mathbb{R}^n$ the choice of $O$ alone separates four different roads (Chapter 6).

**R4** (Why Z8 matters.) "No zero divisors" is not an equation. It is lost by Blur in $\mathbb{Z}/6$ (Chapter 3) and by Build in $\mathbb{R}^2$ (Chapter 6).

**R5** (Old names.) Blur respecting $O$: congruence. One-way blur: compatible preorder. Likeness: tolerance relation. From below: generated congruence. From above: syntactic congruence. Z7: the laws preserved by products, subalgebras and quotients. W4: the first isomorphism theorem. F4: a concept, in the sense of formal concept analysis.

**R6** (What is not a rule.) A rule here may rely on nothing after Chapter 0 (F7 read backwards). So Amplify, a theorem about power sums whose proof uses series and poles (Chapters 17, 20, 33, 34), is not a rule: it lives in Field notes: the edge, Note 2, line A (formerly Z11). The one-way growth blur it uses is Chapter 17's. F8 stays: it is a reading convention and relies on no chapter.

## Exercises

**E1** On $\mathbb{N}$ with the operation $x \mapsto x+1$, find the blur forced by the single gluing $5 \sim 2$. How many bags are there?

**E2** Prove Z4.

**E3** Show that "$x+y=y+x$" is an equation and "$x \neq 0 \Rightarrow$ there is $y$ with $x \times y = 1$" is not. Find where the second is lost in this book.

**E4** Give two different normal forms for the blur of Chapter 2.